

Central Limit Theorem

MATH 213

The Central Limit Theorem

The sampling distributions for a mean or proportion will approach the normal distribution centered on the population mean **as long as some basic criteria about the original data and sampling are met**. These criteria differ depending on context.

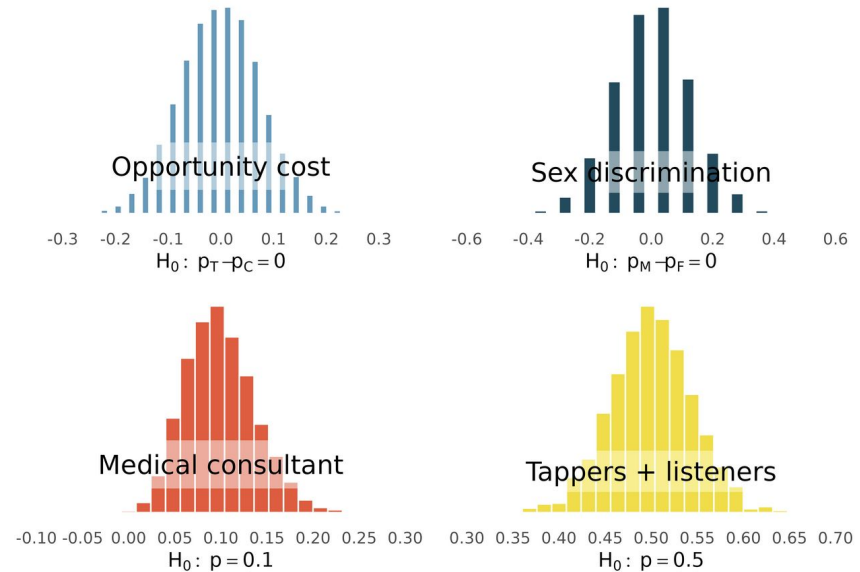


Figure 13.1: The null distribution for each of the four case studies presented previously. Note that the center of each distribution is given by the value of the parameter set in the null hypothesis.

Criteria for one proportion (Chapter 16.2 of your [textbook](#))



Sampling distribution of $\hat{p}$.

The sampling distribution for $\hat{p}$ (the sample proportion) based on a sample of size n from a population with a true proportion p is nearly normal when:

1. The sample's observations are independent, e.g., are from a simple random sample.
2. We expected to see at least 10 successes and 10 failures in the sample, i.e., $np \geq 10$ and $n(1 - p) \geq 10$. This is called the **success-failure condition**.

When these conditions are met, then the sampling distribution of $\hat{p}$ is nearly normal with mean p and standard error of $\hat{p}$ as $SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$.

Warning: independence gets more subtle than this in real life!

Confidence intervals for proportions using the CLT



Mathematical model confidence interval for any confidence level.

If the point estimate follows the normal model with standard error SE , then a confidence interval for the population parameter is

$$\text{point estimate} \pm z^* \times SE$$

where z^* corresponds to the confidence level selected.

We've been using z^* of 2 for 95% confidence and $\frac{2}{3}$ for 50% confidence, but where do those come from? Time to abandon slides to get some real work done.

What about means? Again from your book, [Chapter 19](#)



Central Limit Theorem for the sample mean.

When we collect a sufficiently large sample of n independent observations from a population with mean μ and standard deviation σ , the sampling distribution of $\bar{x}$ will be nearly normal with

$$\text{Mean} = \mu \quad \text{Standard Error (SE)} = \frac{\sigma}{\sqrt{n}}$$

But what does “sufficiently large” mean?

- **Independence.** The sample observations must be independent. The most common way to satisfy this condition is when the sample is a simple random sample from the population. If the data come from a random process, analogous to rolling a die, this would also satisfy the independence condition.
- **Normality.** When a sample is small, we also require that the sample observations come from a normally distributed population. We can relax this condition more and more for larger and larger sample sizes. This condition is vague, making it difficult to evaluate, so next we introduce a couple rules of thumb to checking it.

Additional restriction: we're almost always sampling *without replacement* in real life, which *technically* isn't independent sampling! But this isn't a problem as long as the sample is **SMALLER** than 10% of the population.



General rule for performing the normality check.

There is no perfect way to check the normality condition, so instead we use two general rules based on the number and magnitude of extreme observations. Note, it often takes practice to get a sense for whether a normal approximation is appropriate.

- Small n : If the sample size n is small and there are **no clear outliers** in the data, then we typically assume the data come from a nearly normal distribution to satisfy the condition.
- Large n : If the sample size n is large and there are no **particularly extreme** outliers, then we typically assume the sampling distribution of $\bar{x}$ is nearly normal, even if the underlying distribution of individual observations is not.

Some guidelines for determining whether n is considered small or large are as follows: slight skew is okay for sample sizes of 15, moderate skew for sample sizes of 30, and strong skew for sample sizes of 60.

Bottom line: look at your data before you analyze it!

Confidence intervals for means using the CLT

We'd like to use our z^* trick again, but there turns out to be a very sneaky consequence of our approximation in the SE formula

$$SE = \frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}}$$

that makes it not work as well for means as for proportions

Luckily, a plucky Guinness employee came to the rescue.